

Planning of Available Resources Considering Ergonomics Under Deterministic Highly Variable Demand

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Abstract - In this paper, a method for hybrid short- to long-term planning of available resources for operations is presented, which is based on a known or deterministically forecasted but highly variable demand. The method considers quantitative measures such as the performance and the availability of resources, ergonomically relevant KPI and ultimately process costs in order to serve as a pragmatic planning tool for operations managers in SMEs. Specifically, the method enables exploiting the ergonomic advantages of available flexible automation technology (e.g. AGVs or picking robots), while assuring that these do not represent a capacity bottleneck. After presenting the method along with the necessary assumptions, mainly concerning the availability of data for the calculations, we report a case study that quantifies the impact of throughput variability on the selection of different process alternatives, where different teams of resources are used.

Keywords - Planning, Ergonomics, Resource Allocation, Operations, Service Robots, QIFD

I. INTRODUCTION

Recent statistics on the adoption of robots by sector [1] show that the robot density – the number of multipurpose industrial robots in operation per 10.000 employees – is sensibly higher in the automotive manufacturing sector, typically populated by medium to large size companies, than in any other industrial sector. In the Republic of Korea for instance, about 4,57 times more robots are used in the automotive sector than in the general manufacturing sector, in Germany about 6,08 times more and in Japan 5,15 times. The high robot density in the automotive industry is related to the high level of product and process standardization aimed at reaching the highest throughput and quality with the minimum costs. In such automation-friendly environments, the cost-benefit ratio of employing robotics [2] is positive, because of three conjoined effects: (1) increase in throughput (capacity), (2) decrease in costs and (3) improvement in product quality [3]. However, the achievement of a higher robot density also in sectors with lower standardization level becomes increasingly urgent. This is especially true in industrialized countries, where quality of life is relatively high, unemployment rate is low and population is ageing, as these concurring factors made it increasingly hard to find labor willing to take over ergonomically hard, monotonous, low-paid jobs [4]. Additionally, the total productivity – the amount of goods and services that a group of workers produces in a given

amount of time – is destined to decrease in ageing countries. Studies in the field of demographic change [5] show that the EU is in average headed to a yearly loss of productivity estimated at 0,2%, which is solely due to its aging population phenomenon. Amidst several corrective actions pursuable at best through politics, the use of automation could further support the containment of the productivity loss [6]. However, companies and sectors with low robot density normally cannot achieve high product and process standardization and are therefore caught in the challenge of solving the trade-off among necessary flexibility of the technology, its consequent (normally high) costs and its resulting (normally low) performances [2]. The requirement of technological flexibility typical of SMEs towards automation, namely the capability of the automation to execute different tasks (e.g. handling of heterogeneous goods) [7], arises from the necessity of managing the risk of being overly dependent on customers and on their product and process-specific requirements. Should requirements for the products to be processed change, as they often do in the current dynamic and ever disrupted business environment, investments made by SMEs should be re-usable, ideally with little required adaptation. Along with improved robotic skills empowering automation towards a wider range of application, the last decade of research led also to advancement in control and management systems for pool of resources, both autonomous and human. One such instance are advanced business process management approaches, which bring together both relational process structures and semantic relationships, enabling the coordination of complex interactions of interrelated processes to be accomplished by different types of resources [8]. Advanced technological skills for robotics, in conjunction with progressed process management tools, increasingly enable the quick design and installation of complex system of systems (SoS) via modular approach, as well as their successive adaptation, potentially necessary should process and product requirements change. However, while these progresses providing for high degrees of technological flexibility previously unavailable are undeniable, such potential still struggles to find application in SMEs, largely because of the lack of operative tools enabling the use of flexible technologies. In other words, after the design of a flexible SoS meeting the requirements of a set of potential users in different SME sectors, a barrier to the adoption still exists due to the lack

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of operative tools enabling for instance the planning of operations. The mentioned technological and process flexibility [7] enables the theoretical dynamic assignment of tasks to compatible resources of all kinds (i.e. automated and human); however, operations managers in SMEs still have to solve an apparently more trivial problem of medium to long-term planning. While automated resources, if purchased with the promise of flexibility, can be considered as always available in the planning of operations for the medium to long-term, the availability of human resources must be planned and organized in shifts on the basis of a forecasted demand. Only doing so the amount of human resources available at operation time can be levelled, at least approximately, with the necessary capacity. Trivial problems such as the planning of holiday leaves, of additional capacity in peak periods or simply the calculation of human resources necessary during night shifts, become less trivial when flexible automated resources are taken into account. These automated resources, being compatible with at least some tasks otherwise manually executed, are able to relieve operators e.g. from some ergonomically hard activities: as such, they should be exploited in favor of ergonomics. However, as argued later in the paper, during peak period fully exploiting automated resources may result in bottlenecks to productivity, easily solvable by planning the availability of additional human resources. With the increasing of possibilities and variables for usually trivial medium- and long-term planning task, SMEs managers are overwhelmed by the apparent complexity of the planning, which raises the hurdle to the adoption of flexible technologies, resulting in low robot density.

In this paper, we address this problem by presenting a method implemented in a decisional tool to be provided to SMEs operation managers. The method considers in input a deterministic forecasted amount of necessary capacity, depending, for instance, on the period of the day or on the seasonality of demand. Then, analyzing and evaluating different pre-determined alternatives for the execution of necessary processes on the basis of performance, ergonomics and costs, the method delivers the SME operation manager the number of resources of each type, including therefore the number of operators, that are needed on a shift basis. We highlight that the purpose of this tool is not to dimension the logistic system, nor to analyze potential factory layout adaptations (covered by guidelines as “VDI 2498 Procedure in material flow planning”[9]). Also considerations about the analysis of the investment, such as the breakeven analysis for the analytical planning of the material flow system with regard to the maximum performance to be achieved in operation [10] and the Extended Profitability Appraisal (EPA) procedure by Bullinger [11] are not in the intended scope. Finally, the presented method is also not suited for a real-time planning or task allocation. The context of application of the presented method is that of an existing system, with determined processes, determined available resources and an available forecasted demand, which could deterministically fluctuate. In this context, we evaluate

different configurations of teams of resources for the fulfillment of the forecasted demand, thereby analyzing their performance, ergonomics and costs in order to provide the operations manager with a short- to long term resource plan.

II. RELATED WORK

Before introducing the method, a brief overview on the state of the art is given in order to define: 1) the dimension of flexibility addressed, 2) frame the problem within the acknowledged taxonomy, 3) identify the relevant parameters that should be taken into account during the short- to long-term planning of resources and 4) briefly introduce the available approaches for the optimization, scheduling and planning of complex systems.

1) *Process Flexibility*

Fostered by the technological advances and enhanced versatility, flexible service robots are becoming increasingly popular. Only in 2018 the total number of professional service robots sold rose by 61% to more than 271 000 units, up from roughly 168 000 in 2017 [12]. These flexible technologies, coordinated by a super-ordinated business process management technology [8] allow the so-called process flexibility dimension which directly impacts the variety of processes, tasks and resources a complex SoS is able of coordinating and executing [7]. The process flexibility could be further deployed into subcategories of which the most relevant to this work are (1) the resources allocation and (2) the human-robot-interaction. The first one concerns the dynamic task allocation prioritizing resources according to specific constraints and adapting to new processes/circumstances [13], while the second considers the role of humans within the process execution and their interaction with automation. Depending on the complexity of the task, on the required performance, on the available resources and even on the time of the day, different resources (human or automation) might be allocated to different sub-tasks to seek an overall better efficiency [7].

2) *Task Allocation in Multi-Robot Systems*

When analyzing the problem proposed in this paper, the framework proposed by Gerkey and M.J. Mataric [14] provides a structured approach to describe the multi robot task allocation (MRTA) problem. According to the proposed taxonomy, the MRTA problem addressed in this paper falls within the category of MT-SR/MR-TA. The addressed problem concerns a system, where: (1) robots are able to perform different and multiple tasks (MT e.g. transportation and handling), (2) a task could be performed exactly by a single robot (SR), or it could require (3) multiple robots (MR). Last but not least, thanks to the information available, it is possible to (4) assign the tasks over an extended period of time (TA). Moreover, within the context proposed, the human-robot interaction should also be considered, thereby extending the scope of the classic MRTA problem. As already described, in a flexible and integrated SoS a set of hybrid resources (operators and robots) are able to work together or independently,

according to changing requirements dictated by external circumstances. Such resources could be flexibly organized in a combination of different teams or individual entities, in order to best achieve the target at hand, which may vary in terms of (1) kind and (2) amount of tasks to be performed within the available (3) time horizon.

3) *Relevant parameters for the task allocation*

A method enabling the efficient use of flexible technologies should provide the planner with a solution achieved by finding a good compromise among conflicting goals of high performance and low costs [15]–[17]. While in the past companies focused essentially on achieving capacity- and utilization-related targets, nowadays the focus has shifted on market-oriented targets, thus increasing the need for flexibility in order to enable a fast reaction to volatile demands [17]. Within this scenario, ergonomics increasingly gained importance, becoming a fundamental driver for process design and resource configuration [18], [19]; this is mostly due to an increased sensibility and awareness of the impact of ergonomics on productivity, employee satisfaction and failure rate [20] and to the necessity to comply with upcoming regulations and guidelines. When including the human factor in a decision support method both cognitive and physical factors should be considered. However, the majority of methods for ergonomics developed both in production and logistics areas aims at assessing physical ergonomics risk and total workload allowed within a defined timeframe (e.g. one shift) [21]. In terms of physical ergonomics of logistic tasks, lifting/holding/carrying and pushing/pulling of loads have to be considered. A separate consideration is possible for instance by using a key indicator method of the Federal Institute for Occupational Safety and Health of Germany [22], [23], ISO 11228 or Snook's tables [24] which show maximum acceptable weights and forces for a variety of tasks. Depending on the method, the calculated ergonomic risk or maximum (total) weight depends on both task and worker related factors such as distance, posture, gender, duration, additional resources etc. When evaluating the task ergonomic risk, first the task at hand should be deployed into elementary tasks, then for each one of them, thanks to methods such as Ergonomic Assessment Worksheet or MTM-analysis orientated metabolic cost calculation [25], the punctual ergonomic risk could be estimated. The overall ergonomic risk could be calculated, working backwards, combining the obtained values. These evidences support the statement that a method for planning flexible resources should be taking into account not only performance- (e.g. throughput) and convenience- related (e.g. cost) parameters, but also ergonomics, in first approximation associated with physical factors.

4) *Long-term approaches for the optimization, scheduling and planning of complex systems*

Complex systems where service robots operate along with machines are often the result of the requirements set by dynamic production or assembly environments. In such environments, usually populated by AGVS (Automated Guided Vehicle Systems), approaches based on the simultaneous scheduling are gaining importance since they

bear the potential to improve the overall efficiency of the system by considering production and logistics resources simultaneously for machine and transport system planning and optimization [25]. These elaborate scheduling and planning approaches requires multiple interlinked algorithms and often an extended IT infrastructure, able to guarantee real time data, not always available. Few of the approaches currently developed are actually adopted in industrial environments or demonstrated at lab scale. For instance, combining the method of the Extended Profitability Appraisal (EPA) to determine the most effective work system for order fulfilment with an autonomous control method involving principles of decentralized and target-oriented decision-making it is possible to achieve target-optimized process executions, as proposed and demonstrated by Schuhmacher and Hummel [26]. However, none of the methods found in the literature, considers the aspects of human-centered and human-robot-collaboration with limited expenditure of time for a fast and low effort planning of logistics system.

In the following section, the method we propose is introduced highlighting how the identified relevant parameters are considered in order to achieve the sought solution.

III. METHODOLOGY

In this section, we introduce a method for hybrid short-, medium- and long-term planning of resources to be deployed for operations, which is implemented in an intuitive spreadsheet tool, usable by operations managers. The main purpose is enabling the correct planning of the hybrid resources (that is, human and automated) considering their available capacity, their ergonomic requirements and ultimately their cost. We hereby proceed by listing the assumptions and explaining the method.

A. *Assumptions*

We made two fundamental assumptions about the availability of data enabling the application of the method, namely that (A) the needed capacity (forecasted or actual) is available as well as (B) a quantitative description of the potentially available resources. The main assumption of the method is that (A) data concerning the tasks to be assigned to the resources are *all* available at the time of planning in an aggregate fashion. According to the nature of such data, different time horizons for the planning of resources are possible. If (1) punctual historical data concerning the exact demand for capacity in previous years are available, then the planning method yields a medium- to long-term plan that can be particularly useful to decide which resources should be engaged in operations during peak (e.g. Amazon Prime days, black Friday, cyber Monday, before Christmas etc.) or during slow seasons (e.g. night shifts in summer holiday season). In case (2) the context allows the assignment and accomplishment of tasks on the basis of a long lead-time orders and resources can be booked in advanced on the basis of the actual demand (not the demand forecasted by means of historical data as in the

previous case), then the method can be used for short-term planning of shifts for the next days or weeks (depending on the lead-time and actual deadline for the received orders). In both cases however, the method assumes and takes in input a “frozen” demand for capacity - forecasted in case (1) or actual in case (2) - and not a real-time demand. This is why we highlight that the method does not address the real-time planning of resources. A second assumption necessary to apply the method is that (B) databases are available (e.g. in the Enterprise Resource Planning software, in a Plan For Every Part or in a dedicated database) describing the potentially available resources and the products object of the demand (e.g. parts to be delivered or assemblies) in their numerical characteristics. Example of such numerical characteristics for resources are performance, ergonomics thresholds, operation time etc. For parts and products object of the demand, examples of numerical characteristics are number of parts in each unit-load, weight of a single part, location of a part in a warehouse, or sub-assembly buffer, assembly time etc. Such databases are necessary to calculate indicators of capacity (e.g. identification of bottleneck, or utilization), ergonomics and costs that are used to make the planning.

B. The method for hybrid short- to long-term planning of resources

The presented method for the short- to long-term planning of resources is made of three major steps: (1) collection and aggregation of data concerning the demand of capacity, (2) consultation of the part and sub-assembly database for relevant information and (3) evaluation of best team of resources considering performance, ergonomics and costs. The first step, where (1) data concerning the expected or actual demand for capacity are collected, represents the main input of the method: changes in the aggregated value of needed capacity can yield sensibly different result for the method, as we show in the case-study in next section. Once the data concerning the capacity are collected and deterministically described, they are translated in secondary indicators by means of (2) consultation of a parts and sub-assembly database (e.g. a Plan For Every Part). This enables the calculation, for the capacity calculated in the first step, of indicators such as total weight for the order, total time of assembly, total distance to cover, number of picks etc. In other words, in this second step the demand is translated in planning-relevant quantities by means of internally available information, depending on the experience and on previously conducted investigations (e.g. MTM analysis for time of assembly, optimization of delivery routes for time and distance of delivery etc.). Once the demand for capacity is translated in planning-relevant indicators, a (3) calculation is conducted in order to find out which teams of resources suits best the demand considering performance, ergonomics and costs factors. This step can be divided itself in three sub-steps to be executed sequentially. Each sub-step takes in input information from external sources, such as databases, norms or previously applied method for the decision-making. Starting from a

pool of compatible resources organized in (homogenous or hybrid) teams, which were previously investigated with other methods (e.g. the Quality Interaction Function Deployment QIFD [3]) and that are therefore an input for the presented method, in each step the pool of possible candidates for the allocation of tasks is re-assessed according to different criteria. In the first sub-step, (3.1) the comprehensive pool of initially available and compatible resources is vet for the requirement of capacity calculated in step 2. Those teams of resources that cannot fulfill the minimum required capacity in the available time of operations are sorted out of the pool of possible candidates. The ergonomics requirements of the remaining candidates are then (3.2) crossed-checked with the ergonomic load foreseen by the demanded capacity, with the objective of eliminating those teams of resources, whose ergonomic load exceeds the normed values. A necessary input for this second step are norms for ergonomics, both external (laws) and internal (e.g. negotiated with unions), indicating threshold values. Examples of external norms embedding usable thresholds are the NIOSH procedures (National Institute for Occupational Safety and Health) or the “Ergonomic Assessment Worksheet (EAWS)” or CEN- and ISO-norms for material handling processes (e.g. lifting, holding, carrying and pushing, pulling). Finally, among the remaining candidates (3.3) the team with lowest operative costs is chosen for the deployment in the period of inquiry (a shift, or a peak week etc.). This sub-step takes in input previously assessed costs of operations measured for instance per time unit or per part/order and then up-scaled to the actual demand investigated in step 1.

Summarizing, the method implements a cascade logic to filter out teams of resources that are not suitable to fulfill the necessary capacity, or that would be overloaded from an ergonomic standpoint. Among the remaining teams of resources, the cheapest for operational costs is selected to fulfill the demand. The method not only enables the identification of the best team of resources to fulfill the demand, but it can also flag the lack of suitable resources. This happens if, in the sub steps 3.1 or 3.2 all teams of resources are sorted out, since no team respects both capacity and ergonomics constraint. In this case, it is also calculated for each excluded team, how many additional resources would be necessary in order to fulfill the constraints.

IV. CASE STUDY

The case study presented in this section is inspired by the demonstration scenario put in place to present the results of a EU and German co-funded project ZAFH - Intralogistik [27]. The scenario replicates the assembly workshop of a SME that assembles scooters. The process object of study is the logistic process of commissioning and transportation of parts to an assembly station (point of use). Given customer orders, the parts needed for the assembly are retrieved from a warehouse and delivered to the point of use where they are required. Thanks to flexible

technologies at play to the business process management tool available to the company, the commissioning and transportation processes can take place following three different process configurations. The first option is (1) fully manual, where operators take over both the commissioning and transportation tasks, without the help of any service robot or automated equipment. The second option is a (2) manual process supported by autonomous transport robots, where an operator performs the commissioning activity and the transport activity is taken over by transport robots. A third and last option is (3) the combined manual and autonomous commissioning with automated transport, where all the suitable items are commissioned by autonomous picking robots supported by transport robots, while the non-suitable items are manually commissioned and handed over to the transport robot to be delivered to the point of use. The goal of the manager is to decide on a short-term horizon of one shift the process configuration that allows the fully exploitation of the potential offered by the flexible system. We applied the method presented in the previous section to the described scenario, whose procedure is illustrated in the flow chart in Fig. 1. Assuming that all the information and technical data concerning available technologies and allowed process configuration are known, in the first method step (STEP 1) the customer demand, is aggregated, so that the overall capacity necessary to fulfill commissioning and transportation tasks can be estimated for the given time frame (namely a single shift in the case study). Once this step is completed, the relevant parameters for the planning are calculated (STEP 2). In this second step, we combined the data concerning the aggregated demand with the information included in the plan for every part (PFEP) and the in the enterprise resource planning (ERP). This enables the calculation, based on one entire shift, of parameters such as: number of picks, distance to be covered to deliver the items required and load to be transported [kg*m] and handled [kg]. Once the data have been aggregated in the relevant metrics (STEP2), they are fed to the cascade logic filter (STEP 3), which, as described in the previous section, evaluates the ability of different resources and related process configuration to accomplish the commissioning and transportation tasks, prioritizing capacity and ergonomics over costs. The numerical experiment, conducted under the aforementioned conditions, considers a constant amount of available resources, namely 10 operators, 10 transport robots and 3 autonomous

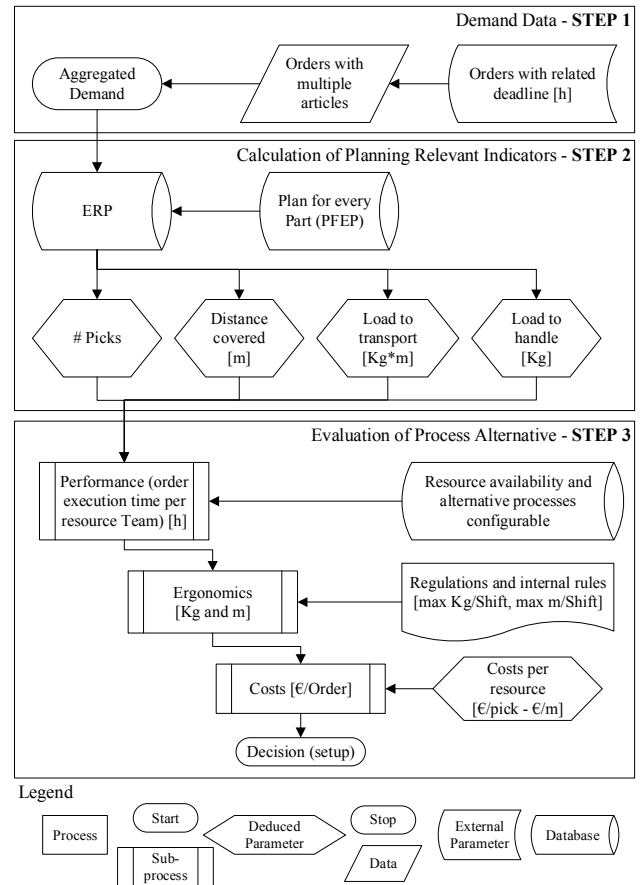


Fig. 1 Method flow-chart

commissioning robots. The experiment aims at configuring the logistic system according to variable throughputs, in terms of items produced, with direct impact on the requirements for commissioning and transportation activities. Assuming (a) that the number of items compatible with the autonomous commissioning technology always accounts for the 20% of the total items to be commissioned and (b) that items are grouped in bundles of five before being transported to the point of use, we repeated the experiment three times changing the variable ordered items, proxy for the required performance. In the first iteration, an (1) average day shift of 8 hours with an order of 1000 items was considered. In the second iteration, we used an (2) average night shift of 8 hours with a consequent order drop of 50% with respect to case (1), to

 TABLE I
 EXPERIMENT RESULTS

	<i>Experiment 1</i>			<i>Experiment 2</i>			<i>Experiment 3</i>		
Scenario	Day shift			Night shift			Peak shift		
Transportation requirement	1000			500			2000		
Process selected	Manual supported by autonomous transport			Combined manual and autonomous commissioning with automated transport			Fully Manual		
Resources	Operator	Transport Robot	Commissioning Robot	Operator	Transport Robot	Commissioning Robot	Operator	Transport Robot	Commissioning Robot
Resources employed	6	6	-	3	5	2	9	-	-
Utilization	0.90	0.98	-	0.72	0.84	0.82	0.93	-	-
Ergonomics (KPI)	0.35	-	-	0.28	-	-	0.41	-	-
Cost per transport [€]	0.34 €			1.22 €			0.65 €		

500 items in the shift. In the last iteration, we considered a (3) peak period shift (e.g. before Christmas or black Friday) with a throughput increase of 100% with respect to case (1), namely where up to 2000 items needed to be commissioned and transported in 8 hours.

V. RESULTS

In Table I, summarizing the results, for each scenario the relevant parameters on which the decision is based are explicitly presented: utilization of capacity, ergonomics and the last controlled variable, costs. It is interesting to notice how the number of resources employed varies according to the selected process alternative and the consequent impact on utilization and ergonomics. In the first two cases (Table I – experiment 1 and 2) service robots could effectively support commissioning processes, responding to changes in throughput. The demanding high throughput requirement considered in the third case (Table I – experiment 3) lead to manual operations, where ergonomics advantages offered by service robots are not exploited, mainly due to the related decrease in performance. As it could be expected, given the lower performance characterizing autonomous technologies in comparison to manual processes, bearing in mind the tasks at hand, during peak periods the preferred processes configuration leans towards fully manual operations, at the expense of the ergonomics KPI (Key Performance Indicator). Within this case study, the ergonomics KPI, which ranges between 0 (best case) and 1 (worst case), has been calculated as the weighted sum of load carried and handled by the operator during a shift, against an acceptable threshold, according to the following formula:

$$E_{KPI} = \frac{l_c}{\max l_c} w_{l_c} + \frac{l_h}{\max l_h} w_{l_h} + \frac{d}{\max d} w_d \quad (1)$$

Where the numerator represents the actual value of load carried l_c , load handled l_h and average distance covered d by an operator within a shift, considering the number of operators necessary to fulfill the performance requirement and according to the specific process configuration. The corresponding maximum thresholds defined for each parameter are in the denominator. The obtained ratios are then weighted by the factors w_i according to their nature and the company policies ($\sum_i w_i = 1$).

VI. DISCUSSION AND OUTLOOK

As already mentioned, the purpose of the developed method is not to dimension the systems, nor to steer investment decisions in innovative technologies, both extremely important aspects regarding the long-term planning. Conversely, the presented method assumes that the resources are already available (therefore dimensioned and purchased/hired) and takes in input their parameters. Precisely the input parameters play a delicate role in achieving a reliable result. As highlighted in section III, the main assumptions on which the method is based concern the availability of information about the needed capacity

and the technical parameters describing the potentially available resources. Especially when considering the variegated scenario of SMEs, the nature and the quality of information may substantially vary. In scenarios where the needed information is not available or it is outdated a considerable effort should be put into the set-up phase where inputs are collected and the IT system, where available, interfaced with the intuitive spreadsheet tool developed. Procedures to guarantee that the input data are accurate and always up-to-date should be contemplated in order to ensure the reliability of the results.

While the proposed method does not address the real-time planning of resources, it allows operations manager to consider ergonomics as a driver in the process configuration. As mentioned, at this stage of method development, only physical factors are considered when calculating the overall ergonomic KPI. Further research should aim, on the one hand, at integrating the cognitive effort in the calculation of the ergonomic KPI and, on the other hand, at developing a dynamic scheduler, which allows the real time task allocation, minimizing the deviation between short- to long term resource plan and actual resource exploitation, according to performance, ergonomics and cost indicators.

VII. CONCLUSIONS

In this paper, a method for hybrid short- to long-term planning of available resources for operations considering deterministic but highly variable demand was presented. Considering a large set of parameters, such as the performance and the availability of resources, as well as several ergonomically relevant KPI, the method serves as a pragmatic planning tool for operations managers. Specifically, the method enables exploiting the ergonomic advantages of available flexible automation technology (e.g. AGVs or picking robots), while assuring that these do not represent a capacity bottleneck. If a capacity bottleneck is identified, a different process alternative is chosen, in which operators can work in parallel to share the ergonomic load. The presented case study quantifies the impact of throughput variability on the selection of different process alternatives, where different teams of resources are used. Three different capacity requirements were used for the calculation, exemplifying a regular, a reduced and a peak operational period. The method highlighted the possibility of optimizing ergonomics only in with regular or reduced operations, while in peak periods the high performance of operators is still required to fulfill the demanded capacity. The simplicity of the method should guarantee its usability in the context of SMEs, thereby helping operations manager planning the usage of available resources by analyzing and evaluating different pre-determined processes alternatives, considering performance, ergonomics and costs on a fixed time horizon, for which the demand is known or it can be deterministically forecasted.

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